DETECTION OF HELICOBACTER PYLORI BACTERIA

Automated Object Detection Pipeline for Clinical Pathology

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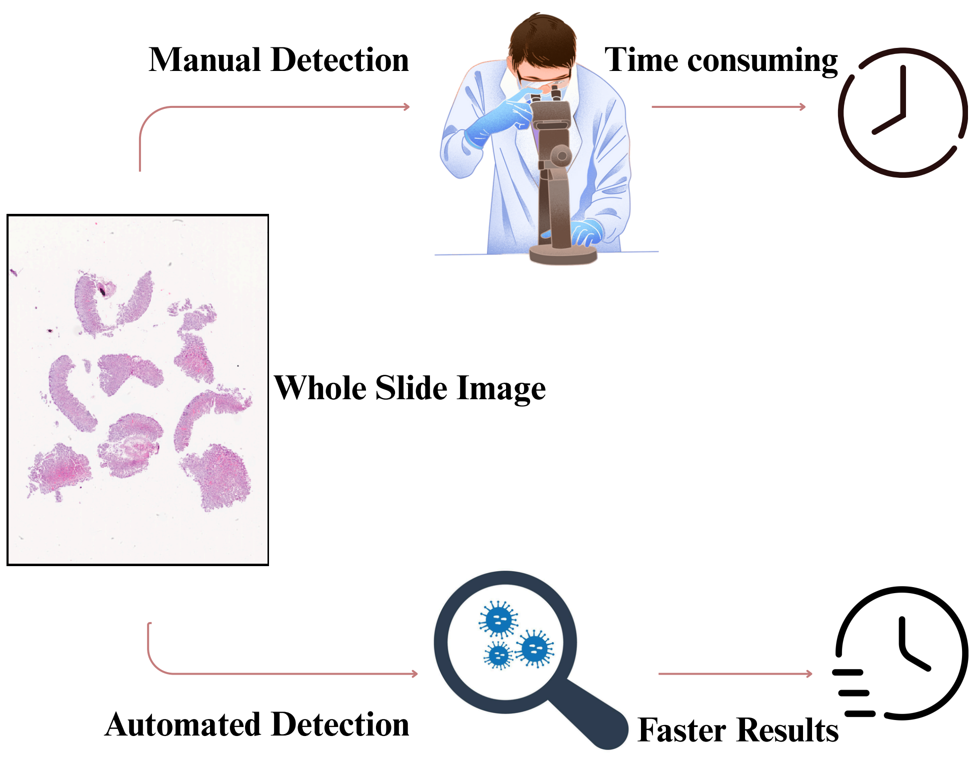
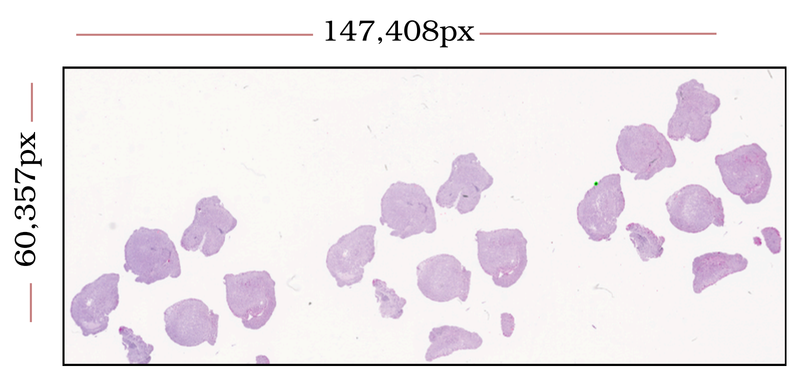
1. PROBLEM AND MOTIVATION

H. pylori, a carcinogenic bacterium casuing ulcers and cancers. Current manual microscopy approaches are slow, inconsistent results, and exhausting for pathologists. Thus, an automated detection system is required to improve diagnostic efficiency and accuracy.

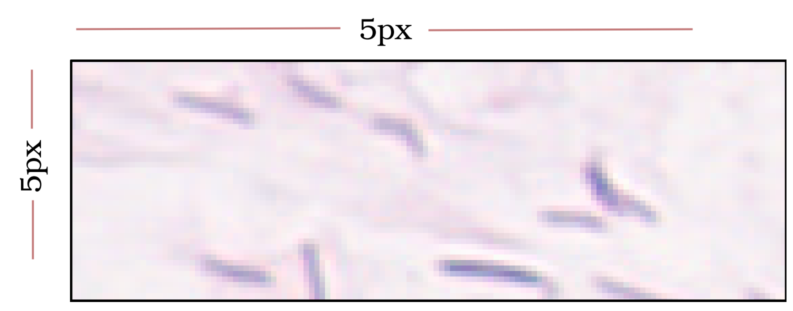


2. KEY CHALLENGES

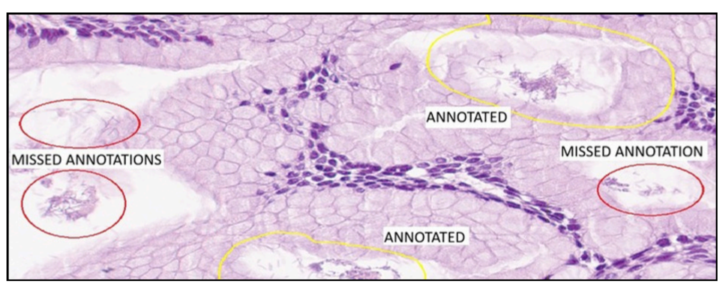
- Massive gigapixel slide image



- Few pixel sized bacteria



- Bacteria not spotted during annotation



3. WHAT MAKES OUR APPROACH UNIQUE?

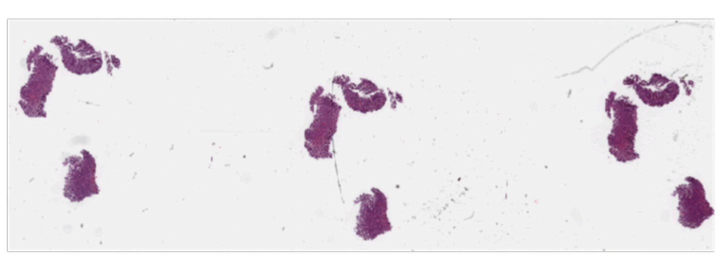
Active learning for missed annotations
Annotations for additional ground truth

Hierarchical detection pipeline
Handle multi-scale detection from WSI to patch level

Dual model approach
YOLO and R-CNN

4. METHODOLOGY

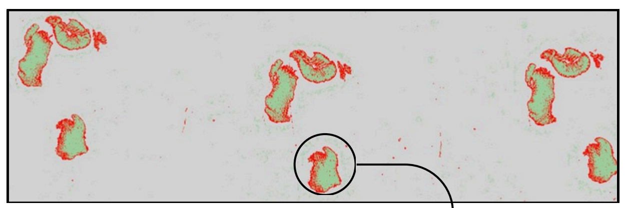
WHOLE SLIDE IMAGE - WSI



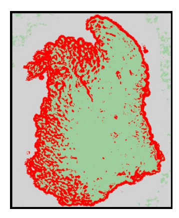
WSI PREPROCESSING

Gland segmentation

Patch extraction

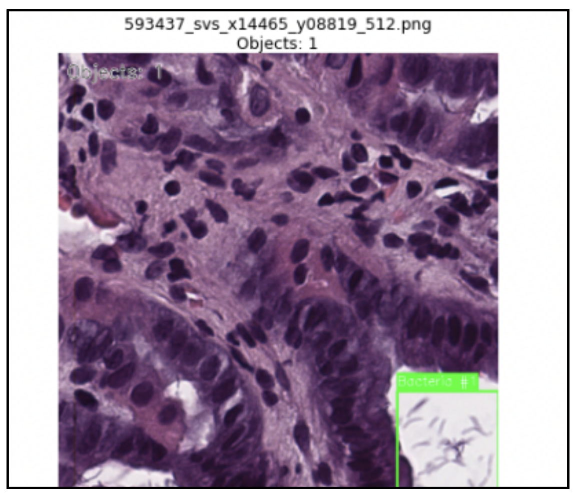


Close up image of gland segmentation with prone regions marked in red, green are stroma and rest are background which can be discarded.



OBJECT DETECTION MODEL

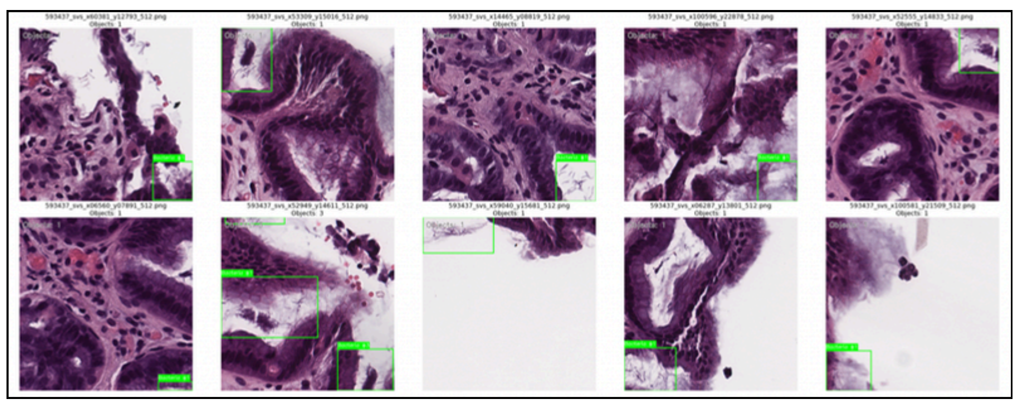
Identify bacteria regions with bounding box



ACTIVE LEARNING STRATEGY

OUTPUT

High confidence detected regions (shown 10)



TECHNICAL DETAILS

Dataset overview

22 gastric biopsy WSI from LABOKILN
Train:15, Test:7

Region and patch processing

Tool: QuPath
Sliding window of 512 x 512
20% patch overlay
Convert prone regions to patches
Maintain context for bacteria

Data augmentation

For upscaling positive images to avoid data imbalance.
HSV adjustment (hue, sat, value)
Geometric transforms (flip, rotate)

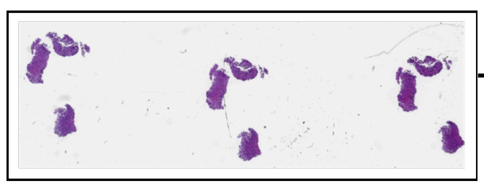
Object detection

Dual model approach
Model: YOLOv8 and R-CNN
Validation: precision, recall, IoU

Active learning strategy

Train model → Identify high-confidence predictions → Expert review for verification → Add verified patches → Retrain.
Expanded dataset from ~500 to ~1000 positive patches.

5. EXPECTED OUTCOME



YOLO
Faster screening in seconds
High confidence

Bacteria Detected

Yes ✓

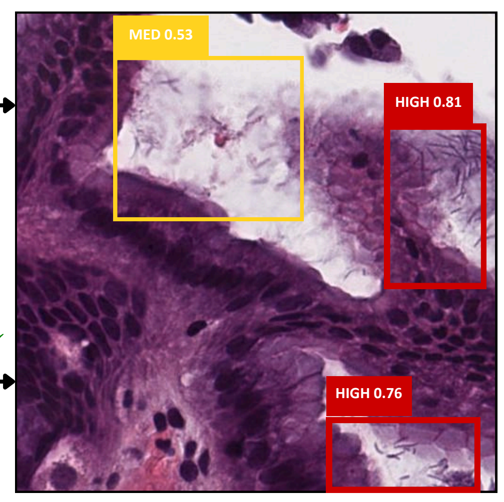
No ✗

R-CNN
Double check in minutes
Deep verification

Yes ✓

No ✗

Confirmed negative
No bacteria found (Animal classified healthy)



We employ a two-stage detection pipeline: YOLO serves as the primary model for rapid bacterial detection in slide images. For slides that YOLO classifies as negative, we apply R-CNN as a secondary verification step to confirm the absence of bacteria.

6. REFERENCES

- Klein et al. (2020). "Deep learning for sensitive detection of Helicobacter Pylori in gastric biopsies". BMC Gastroenterology 20.1, p. 417.
- Yang et al. (2020). "Detecting helicobacter pylori in whole slide images via weakly supervised multi-task learning". Multimedia Tools and Applications 79.35-36, pp. 26787–26815.